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Ayush Agarwal

Technical Skills:

- **Languages:** Python, TypeScript
- **AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, RAG
- **Libraries/Frameworks:** Scikit-learn, Pandas, NumPy, NLTK, spaCy, LangChain
- **Models/Techniques:** LDA, Random Forest, Decision Tree, AdaBoost, KNN, TF-IDF, NER, Embeddings (MPNet)
- **Retrieval/Graph:** FAISS, BM25, Knowledge Graphs, PageRank, RRF, Vector Databases
- **Backend:** FastAPI, REST APIs, WebSockets, SSE, Async Programming
- **Frontend:** React, Streamlit, Tailwind CSS
- **Databases:** SQLite (FTS5), Azure CosmosDB, Azure Blob Storage
- **Cloud/DevOps:** Microsoft Azure, Docker
- **AI Systems:** Multi-Agent Systems, Function Calling, MCP, Pydantic, Instructor, Prompt Engineering
- **Tools:** Tesseract OCR, pdfplumber, pdf2image, PIL, NetworkX, PyVis, RapidFuzz, Joblib
- **Auth/Security:** OAuth 2.0, MSAL
- **Monitoring:** OpenTelemetry, Arize Phoenix

Certification:

- Prompt Engineering for Developers – DeepLearning.AI (Andrew Ng)
- Machine Learning – NPTEL
- Introduction to Large Language Models – Google Cloud
- Generative AI – Google Cloud

EDUCATION			
Board	Tenure	Educational institution	CGPA/Percentage
B. Tech (CSE) Core	Sep 2023 - Ongoing	VIT Bhopal University	8.52/10
Class XII	May 2022	D.A.V Public School, Ranchi	88.0%
Class X	May 2020	D.A.V Nandraj Public School, Ranchi	91.0%

ACADEMIC PROJECTS	
End-to-End AI System	Hybrid Legal RAG: Sparse–Dense–Graph Retrieval System • Engineered a production-grade retrieval-augmented system combining BM25 (FTS5), dense embeddings (MPNet + FAISS), and graph-based ranking (PageRank) • Implemented Reciprocal Rank Fusion (RRF) to unify multiple retrieval modalities for improved ranking quality • Processed and indexed 26K+ documents, enabling high-precision semantic and keyword-based search • Achieved 82.4% Top-1 accuracy and NDCG@10 of 0.89, outperforming single-modality baselines • Built a FastAPI backend supporting search, conversational querying, and structured summarization • Conducted ablation studies to validate robustness under high-noise conditions • Containerized deployment using Docker with memory-efficient optimizations
Full-Stack LLM Application	PolarBrief AI – Legal Argument Analyzer • Developed an AI-powered document analysis tool to extract, classify, and score arguments from unstructured PDFs • Integrated LLMs via LangChain for summarization, heading generation, and structured scoring • Implemented argument polarity classification (Pro/Con/Neutral) with weighted ranking using TF-IDF + LLM outputs • Designed a hybrid OCR + text extraction pipeline using Tesseract and PDF parsers • Generated structured outputs in JSON/PDF formats with citation-aware summaries • Built an interactive Streamlit interface for visualization and report generation

NLP Pipeline System	Knowledge Graph Pipeline for Document Intelligence - Built an end-to-end knowledge graph extraction pipeline from unstructured documents using NLP and LLMs - Implemented multi-stage processing: OCR, cleaning, chunking, entity-relation extraction, and validation - Designed deterministic ID mapping and cross-chunk entity resolution using fuzzy matching - Normalized entities and relations to construct a clean, canonical knowledge graph - Visualized graphs using NetworkX and PyVis for interactive exploration - Exported structured graph data for downstream analytics
Machine Learning Pipeline	WESAD Multimodal Emotion Recognition - Implemented and compared ML models (LDA, Random Forest, Decision Tree, AdaBoost, KNN) for emotion classification - Processed multimodal physiological signals (ECG, EDA, EMG, Respiration, Temperature, BVP, ACC) - Designed experiments to evaluate sensor modality combinations (chest vs wrist vs combined) - Built pipelines for cross-validation, accuracy, and F1-score evaluation - Generated detailed reports identifying optimal modality configurations

INTERNSHIP	
AI/ML Intern – Aug 2025 – Present	▪ Docu3C Technologies: Reported to: Daniel Ingitaraj K, Head of Product Engineering - Developed and deployed AI/LLM-based applications for financial document intelligence using Azure AI ecosystem - Built end-to-end machine learning and LLM pipelines with structured outputs using Pydantic and schema validation - Designed multi-agent systems and tool-calling workflows using MCP (Model Context Protocol) - Implemented LLM orchestration pipelines using Instructor for structured generation - Built scalable backend systems with FastAPI, supporting real-time streaming (SSE/WebSockets) and async execution - Developed interactive UI applications (Streamlit/React) for AI-driven workflows and validation systems - Leveraged Azure ML, Cognitive Services, and GPU infrastructure for scalable experimentation and deployment - Maintained production-grade code quality using PEP8 standards, Ruff linting, and modular architecture - Utilized modern tooling (uv) for efficient dependency and environment management - Conducted benchmarking, evaluation, and iterative optimization of AI systems

CO-CURRICULARS	
Coding	▪ Solved 125+ Data Structures & Algorithms problems on LeetCode

EXTRA-CURRICULARS AND ACHIEVEMENTS	
Achievements	▪ Worked on advanced AI/ML and LLM-based systems including RAG pipelines, multi-agent systems, and document intelligence ▪ Gained hands-on experience with Azure AI ecosystem and production deployments ▪ Actively engaged in collaborative development and agile workflows
Responsibilities	▪ Event Lead, Linpack Club, VIT Bhopal University Organized and managed technical events and workshops, coordinating with teams and handling end-to-end execution Led planning and operations, ensuring smooth execution and participant engagement ▪ AI/ML Intern, docu3C Collaborated in building scalable AI systems and participated in team-based engineering workflows

Extracurricular	▪ Volleyball (represented Thunders at Aavahan, VIT)
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ADDITIONAL INFORMATION	
Hobbies	▪ Exploring AI advancements ▪ Volleyball
Languages	▪ English, Hindi